Humor Prevails! - Implementing a Joke Generator into a Conversational System

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Abstract. This paper contains the results of evaluation experiments conducted to investigate if implementation of a pun generator into a non-task oriented talking system improves the latter's performance. We constructed a simple joking conversational system and conducted one user evaluation experiment and two third person evaluation experiments. The results showed that humor does have a positive influence on the dialogue between humans and computers. The implications of this fact and problems that occurred during the research are discussed. We also propose how they can be solved in the future.

Keywords: natural language processing, humor, jokes, Japanese puns, pun generator, non-task oriented conversational system.

1 Introduction

1.1 Computing Humor

Even in modern AI, humor processing is still a heavily neglected domain. Also in NLP (natural language processing), research projects on joke generation and recognition are often said to be not really relevant to the main tendencies of the field – which, in fact, does not seem to be proper, as some experiments proved that humor may have a very positive influence on language using machines. For example, experiments conducted by John Morkes et al. [1] proved, that humor enhances task-oriented dialogues, in CMC (computer-mediated communication) as well as in HCI (human-computer interaction). Also results described in this paper (see below) show, that we do need talking engines that would be able to generate humor.

Providing that jokes are "products" of humor (a commonsense definition), it can be stated, that, considering the NLP methodology, the most "computable" genre of jokes is "puns". They can be found in most existing languages – in some of them, however, puns are easier to create and thus their amount is much bigger than in others. One such language is Japanese, in which puns (called *dajare*) are one of main humor genres. This makes Japanese a perfect surrounding for pun processing research. However,

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although some attempts of constructing pun generating engine have been made, also in Japanese, creating a funny joking conversational system still remains an unfulfilled challenge for NLP researchers.

1.2 Talking for Fun

Nowadays, our interaction partners often are made of silicon and wires – in fact, the relations between us and them often become the subject of scientists' interest. One of the theories (backed up with very robust research) says, that people respond to computers using the same social attitudes and behaviors they apply to other people (SRCT – Social Responses to Communication Technologies theory) [2]. Assuming this to be generally true, if we desire our partners to be able to talk in a free way, also computers should be capable of performing such behaviors. Therefore, construction of systems talking in such non-task oriented manner is supposed to be one of the major goals of today's AI. However, this field is still heavily neglected and requires more research.

Using humor to improve chatbot's performance has already been suggested by Binsted [3], and an attempt of combining joking system JAPE [4] with a talking system Elmo (which, in fact, is not completely non-task oriented system – see reference for details) was made by Loehr [5]. The results were relatively poor, for there was barely any relevance between user's input and system's humorous output. Thus, construction of a funny, humor-equipped talking system still requires much effort and there is very much to be done in this field.

2 Our Two Systems

This research is aimed to investigate if humor really does have a positive influence on non-task oriented conversational system. To do that, we combined our joke generator PUNDA Simple - a simplified version of PUNDA Japanese pun generating system [6] - with a conversational system Modalin created by Higuchi [7]. The combined version of these two systems was named "Pundalin" and used as a main system in our experiment.

The reason for simplifying the pun generating engine was that we wanted to check if humor can improve the human-computer non-task oriented dialogue, in a possibly simple way. If the experiment proved that humor has bad influence on the dialogue or is irrelevant, construction of a more complex system would be pointless. We also wanted to avoid time losses in system's response generation, for in more sophisticated pun generation algorithms, the computation process is very time consuming. Although we agree that this problem has to be solved, in this experiment we assumed that waiting for the system's answer would make user bored, which in turn would probably affect his evaluation of the system.

2.1 Modalin

The baseline system we used in our research was Modalin non-task oriented keyword-based conversational system, developed by Higuchi et al. [7]. The system automatically extracts sets of words related to a conversation topic set freely by a user. The

extraction is based on keywords spotted in user utterances. Than, Goo [8] (in previous version – Google) snippets are used to extract word associations in real time without using earlier prepared resources, such as off-line databases. After the extraction process, the system generates an utterance by adding modality to the retrieved proposition, and verifies the semantic reliability of the proposed sentence. Evaluation experiment showed that over 80% of the extracted word associations were correct. For the details, see reference [7]. Modalin was also used as a "base" for creating a joking system Pundalin – see section 2.3 for details.

2.2 PUNDA Simple

We developed this simple system as a part of PUNDA research project [6], aimed to create a Japanese joking conversational system. PUNDA Simple is based on a simplified version of the algorithm of the main PUNDA system, which, although still under development, at its current state can be used as a pun generating support tool.

Although PUNDA Simple was created for the need of this research, the main part of the algorithm is similar to the one used in the main system.

The algorithm. The PUNDA Simple algorithm consists of two parts: Candidate Selection Algorithm (CaSA) and Sentence Integration Engine (SIE) – see Figure 1.

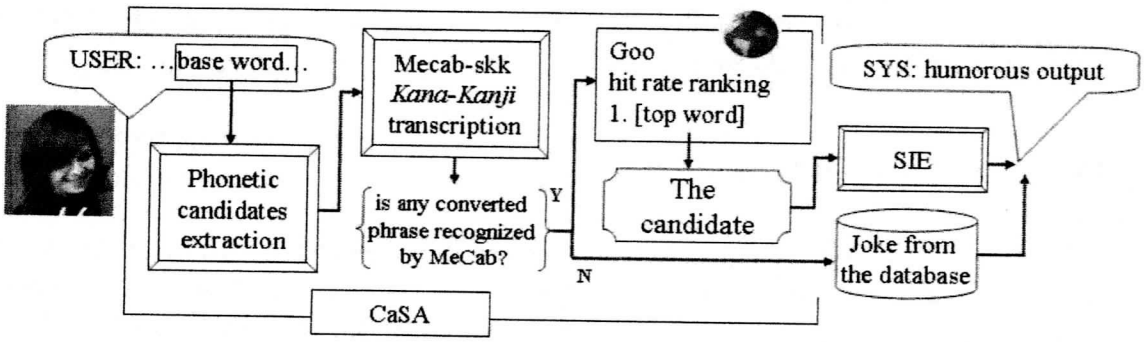


Fig. 1. Algorithm outline for PUNDA Simple joke generating engine

CaSA. In this step, the system generates a candidate for a pun. The input is a sentence, from which a base word for pun (a word that will be transformed into a pun) is selected. The input is analyzed by morphological analyzer MeCab [9], and if any element is recognized as an ordinary noun, it becomes the base word for the pun (a preliminary experiment proved that most of *dajare* base words are ordinary nouns). If no ordinary noun is found, one medium-sized (with medium amount of characters) sequence of characters recognized by MeCab as an independent word is selected randomly. Then, for the base word, pun candidates are generated using 4 generation patterns: homophony, initial mora addition, internal mora addition and final mora addition. For example, for the word *katana* (a Japanese sword), the process goes as follows:

base word: {katana}
candidates:
1. **homophony**: {katana}
2. **initial mora addition**: {*katana } (akatana, ikatana, ukatana...)

3. **final mora addition**: { katana*} (katanaa, katanai, katanau...)
4. **internal mora addition**: {ka*tana}, {kata*na} (kaatana, kaitana, kautana...) (with * meaning one single mora).

The candidates are generated in *Kana* characters (one character = one syllable). In the next step, for each candidate a list of possible *Kanji* (Japanese ideograms) transcription is extracted using MeCab-skkserv *Kana-Kanji* Converter [10]. Then, if any of converted sequence of characters was recognized by the morphological analyzer as an existing word, its hit rate was checked in the Internet. The candidate with highest Goo hit rate was extracted as a pun candidate. For example, for the word *katana* the candidate would be *takatana* (a high shelf).

SIE. In this step, a sentence including the candidate extracted by CaSA is generated. To make system's response more related to the user's input, each sentence that included joke started with the pattern "[base word] *to ieba*" ("Speaking of [base word]"). The following part of the sentence is extracted from KWIC on WEB online Keyword-incontext sentences database [11] - the candidate is used as a query word and the list of sentences including this word is extracted. All sentences that contain non-Japanese characters (such as Latin letters, brackets, numbers etc.) are deleted, as they may be difficult to understand without the context. Then, the shortest sentence with exclamation marks is selected (we assumed that most jokes convey some emotions, and exclamation marks are its visible sign). If no such sentence is found, the shortest sentence that ended with a period is selected. Finally, the part following the candidate is extracted and added as the second part to the system's response – see the example below:

User: - *Natsu wa atsui desu kedo, Toukyou hodo ja nai desu.*(The summer is hot, but not as hot as in Tokyo) [base word: *natsu* (summer), pun candidate: *natsukashii* (emotionally attached; loved)]

System: -*Natsu to ieba, natsukashii Nose de, kyuuyuu to waiwai!* (Speaking of summer, it will be fun to meet some old friends in my beloved Nose! <a town near Osaka>)

If no appropriate sentence is found on KWIC, the response ends with "[candidate] *da yo ne*" ("it's the [candidate], isn't it") – see the example below.

User: -*Ame da kara kaimono iku no wa mendokusai ne.* (It's raining so going shopping is a bad idea) [base word: *ame* (rain), pun candidate: *ame* (a candy)]

System: -*Ame to ieba ame da yo ne.* (Rain reminds me of candies.)

When the candidates list is empty, the system selects one random pun from Sjöbergh's pun database [12] – see the example below.

User: - *Sou, atarashii baito ne.* (Yeah, it's my new part-time job) [base word: *baito* (a part-time job), pun candidate: none]

System's response (from the pun corpus): -*Kureyon wo katte kureyon!* (Buy the pen, man!).

2.3 Pundalin

The two systems described above were merged to create a joking conversational system Pundalin. The rule was very simple – in every third dialogue turn Modalin's output was replaced by PUNDA Simple's joke. In other words, every third user's utterance became PUNDA's input and an appropriate pun for it was generated, using the algorithm described in section 2.2. This method, albeit quite simple, allowed us to check if the usage of humor improved the system's overall quality.

3 Two Evaluation Experiments

To check if humor can enhance the non-task oriented dialogue, we conducted two evaluation experiments, using Modalin as the baseline system and Pundalin as the main, humor-equipped system.

3.1 User's Evaluation

In the first experiment, users were asked to perform a 10-turn dialogue with Modalin, and then with Pundalin. No topic restrictions were made, so that the talk could be as free and human-like as possible. Thus, the utterances variety was quite big – most of them, however, resembled normal (human-like) beginning of conversation, for example: "What did you do yesterday?", "May I ask you a question?" or "It's hot today, isn't it?"

There were 13 participants of the experiment, 11 males and 2 females; all of them were university undergraduate students. After talking with both systems, they were asked to fill a questionnaire about each system's performance. The questions were: A) Do you want to continue the dialogue?; B) Was the system's talk grammatically natural?; C) Was the system's talk semantically natural?; D) Was the system's vocabulary rich?; E) Did you get an impression that the system possesses any knowledge?; F) Did you get an impression that the system was human-like?; G) Do you think the system tried to make the dialogue more funny and interesting? and H) Did you find system's talk interesting and funny?. The answers for questions were given in 5-point scale with some explanations added. Option "I don't know" was also available, however none user did chose it. Each user filled two such questionnaires, one for each system. The final, summarizing question was: "Which system do you think was better?". Although the question may seem too general, it was added on the basis of common sense. Also, thanks to this last question we do not have to rely on detailed questions only, which gives us another interesting point of view in the evaluation experiment.

3.2 Third Person Evaluation

To verify user's assessment, a third person evaluation experiment was conducted. For each Modalin (non-humorous) dialogue from user's experiment, first three turns were selected, and user's third utterance was manually used as an input for PUNDA Simple to generate a humorous response. The reason for shortening the dialogues was that we wanted to check also the "Modalin plus PUNDA" version, for it was much easier to compare it with Modalin-only dialogue (the content was the same, apart from the last

system's response). Therefore, after altering the third system's response, the rest of the dialogue became irrelevant, and for this experiment's purpose it had to be ended after this turn.

Each participant evaluated 1 set, including 3 short dialogues: Modalin only, Modalin plus PUNDA (with the third system's response replaced by PUNDA's joke) and Pundalin. There were 13 sets of dialogues, and each of them was evaluated 5 times. For the dialogues were too short to be evaluated in details, in the questionnaire one general question was: "Which dialogue do you find most interesting and funny?"

The evaluators were 65 university students, 37 male and 28 female. They were not told of the origin of dialogues and apparently did not know that some of evaluated utterances were generated by computer.

4 Results

The results of these two experiments clearly show, that humor enhances non-task oriented dialogue, as in all categories humor-equipped systems received higher scores and were evaluated as better and more interesting than non-humor systems.

4.1 User's Evaluation

11 out of 13 users (84.6%) evaluated Pundalin (humor-equipped system) as better than Modalin (non-humor - equipped) – see Figure 2. This shows that implementation of humor enhanced the conversational system's performance.

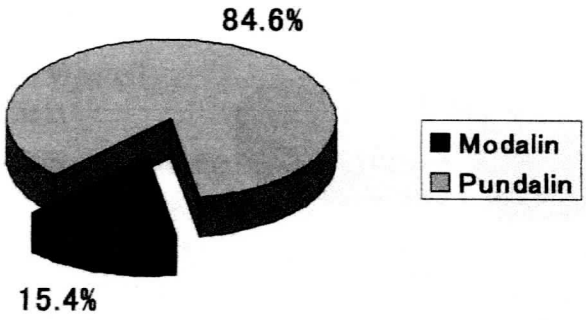


Fig. 2. User's evaluation – results for the question "Which system do you think was better?"

Pundalin received higher scores also in detailed questions (see Table 1). The difference was especially visible in questions G and H, which we consider to be of high relevance in humor research, as they directly address the presence and positive role of humor in the dialogues.

The results of detailed questions evaluation were checked for significance using student t-test (paired). Apart from questions A and B (P value > 0.05), all the results turned out to be statistically significant on 5% level. However, overall results of both systems were relatively lower than they could be. Possible reasons for these are discussed in section 5.

Table 1. User's evaluation – results for Modalin (non-humor equipped system) and Pundalin (humor-equipped system) for detailed questions: A) Do you want to continue the dialogue?; B)Was the system's talk grammatically natural?; C) Was the system's talk semantically natural?; D) Was the system's vocabulary rich?; E) Did you get an impression that the system possesses any knowledge?; F) Did you get an impression that the system was human-like?; G) Do you think the system tried to make the dialogue more funny and interesting? and H) Did you find the system's talk interesting and funny? Answers were given in a 5-point scale.

Questions	A	B	C	D	E	F	G	H
Modalin	2.62	2.15	1.85	2.08	2.15	2.38	1.92	2.46
Pundalin	3.38	2.92	2.69	3.00	2.85	3.31	4.15	4.08
difference	0.76	0.77	0.84	0.92	0.70	0.93	2.23	1.62

4.2 Third Person Evaluation

Also in the second experiment, humor-equipped system scored higher than the non-humorous one. The question asked in this evaluation was: "Which dialogue do you find most interesting and funny?" Evaluators could choose between 3 options: Dialogue 1 (Modalin's first 3 turns), Dialogue 2 (Modalin's first 3 turns with system's third response replaced by PUNDA) and Dialogue 3 (Pundalin's first 3 turns).

Among 65 evaluators, only 10 (15.4%) responded that Dialogue 1 was most interesting and funny. 20 (30.8%) pointed out Dialogue 2 and 35 – Dialogue 3 (53.8%). This means that each of humor containing dialogues received evaluation clearly higher than non-humor dialogue (84.6% as a sum of both humorous dialogues) - see Figure 3.

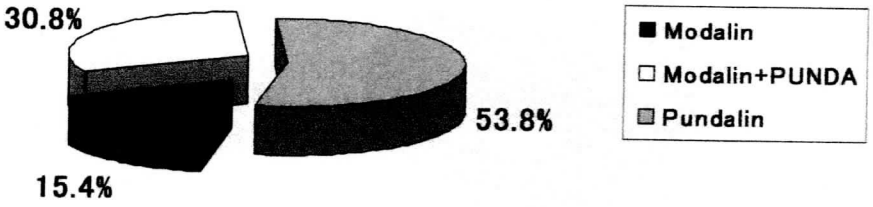


Fig. 3. Third person evaluation - results for the question "Which dialogue do you find most interesting and funny?" (short dialogues)

Although it is clear that both humorous dialogues received much better score, such high score for Pundalin was rather surprising. This is discussed in section **5**.

5 Discussion

Our presumptions turned out to be correct – humor did improve the talking system's performance. In both experiments, Pundalin's dialogues received higher scores than Modalin, in the general evaluation as well as in detailed questionnaire.

As far as the second one is concerned, the most important and relevant for this research are questions G (Do you think the system tried to make the dialogue more

funny and interesting?) and H (Did you find the system's talk interesting and funny?). At these two categories, the differences between the two systems are clearly visible (2.23 point and 1.62 point respectively). This means, that not only did the system try to amuse the user, but the attempts were generally successful.

Also the increase in other categories, such as the will of continuing the dialogue (question A), the possession of knowledge (question E) or system's human-likeness (question F), albeit not directly related to humor, were probably affected by its presence. This also proves that the implementation of joking engine can make talking systems more natural and close to humans.

Remaining categories (questions B, C and D) are all related to system's language abilities. Although it may seem that the presence of humor can distract user's attention from grammatical mistakes or semantic irrelevance, the reason for Pundalin's victory in these aspects probably is that it uses human-created sentences, which obviously are more correct than these generated by Modalin.

Another interesting issue in the experiment results was the the fact that Pundalin scored 23% higher than Modalin+PUNDA in the third person evaluation. The reason for providing the experiment participants also with Modalin+PUNDA dialogue was that we thought it should be easier to tell the difference when comparing it to the Modalin only dialogue (which, apart from the last response, was exactly the same as Modalin+PUNDA). However, the results proved that it does not necessarily have to be true, as more users pointed at Pundalin as the funnier and more interesting system. One possible explanation is that that Pundalin's jokes were funny enough to drag the evaluators' attention and convinced at least some of them that this system was better than two others.

6 Additional Experiment

The results described above encouraged us to convey another experiment, in which third person evaluators would assess the whole-length Modalin and Pundalin dialogues. For few reasons, we expected the results to be slightly lower than the previous ones. First, the joking engine has no context integration algorithm, which is obviously more visible in the long dialogues. Also, reading jokes-including conversations should be less funny than getting a humorous answer from the partner in a conversation (comparing to the users' evaluation).

6.1 The Method

The questionnaires for long dialogues were similar to these used in user's evaluation experiment, with few significant differences. First, the word "system" was changed to "dialogue" or "speaker", so that the participants would not know that some utterances were computer-generated (in the chat logs speakers were marked as "Speaker A" for the user and "Speaker B" for the system). For the same reason, the question F (about human-likeness) was deleted. Also, in questions B, C, D, E and G two sub-sections were added: 1) "Speaker A" and 2) "Speaker B" – so that the dialogue participants

could be evaluated separately. For question H, these two options plus a third one: "overall" – were added. In the end, evaluators had to answer the final question, the same as in the previous experiment - "Which dialogue do you find most interesting and funny?"

The chat logs were divided into 13 sets, each of which included one Modalin and one Pundalin dialogue. Every set was evaluated by 5 participants, what makes a total of 65 evaluators. All of them were university students.

6.2 Results

As we expected, the results for long dialogues were not that good as in previous experiments. However, the overall question's results still show that humorous system is visibly better than the non-humorous one. 45 out of 65 evaluators (69.2%) pointed at Dialogue 2 (Pundalin) as more interesting and funny (see Figure 4).

For the remaining questions, we only take in consideration results for Speaker B (as Speaker A was human, which we do not evaluate here). In all questions, Pundalin dialogues received higher scores than Modalin (see Table 2).

Table 2. Third person evaluation – results for Modalin (non-humor equipped system) and Pundalin (humor-equipped system) for detailed questions: A) Do you want to read the continuation of the dialogue?; B)Was Speaker B's talk grammatically natural?; C) Was the Speaker B's talk semantically natural?; D) Was the Speaker B's vocabulary rich?; E) Did you get an impression that the Speaker B possesses any knowledge?; F) <deleted – see section **6.1**>; G) Do you think the Speaker B tried to make the dialogue more funny and interesting?; H-1) Did you find the dialogue interesting and funny in general? and H-2) Did you find Speaker B's talk interesting and funny? Answers were given in a 5-point scale.

Questions	A	B	C	D	E	F	G	H-1	H-2
Modalin	2.60	1.78	1.48	2.03	1.87	X	2.51	2.88	2.73
Pundalin	2.89	2.09	1.69	2.38	2.13	X	2.91	3.19	3.16
difference	0.29	0.31	0.21	0.35	0.26	X	0.40	0.31	0.43

Although all results showed that Pundalin was better, the student t-test (unpaired[1]) showed that – apart from questions D and G – the results are not statistically significant on 5% level. This issue is discussed in section **6.3**.

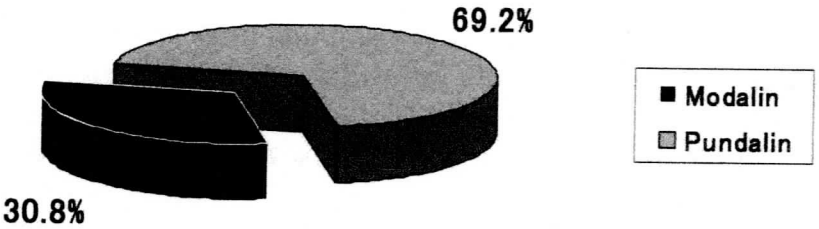


Fig. 4. Third person evaluation - results for the question "Which dialogue do you find most interesting and funny?" (long dialogues)

[1] The reason we decided to conduct unpaired test was that some evaluators choose the "I don't know" answer (also available on the form). Such answers were not taken into consideration when counting points, and this caused differences between numbers of valid answers for each question.

6.3 Discussion

Consistently with previous results, also the third person long dialogues evaluation experiment showed that humor-equipped system is generally better. As we expected, the differences between the two systems were slightly smaller and most of the results are considered not statistically significant (P value > 0.05). However, in one of the most relevant to our research question G ("Do you think the Speaker B tried to make the dialogue more funny and interesting?"), the results were significant (P < 0.05) – this clearly shows, that also third person evaluators at least appreciated Speaker B's (system's) "effort" to make the talk more interesting.

Although most of the results were not statistically significant, in the overall third person evaluation Pundalin still scores much higher than Modalin (69.2% vs. 30.8%). This shows that – even when the differences between dialogues are not statistically significant – the evaluators preferred the humorous ones. Obviously, Pundalin's rich vocabulary as well as grammatical and semantic correctness also had an influence on evaluator's preferences, however, there is a high possibility, that humor is also a factor that "makes things better" here. This, however, is hard to confirm empirically.

We suppose that generally smaller (and less significant) differences in this experiments were caused by the lack of context integration algorithm in pun generator – for a third (non-user) person, jokes generated by the system may seem irrelevant to the topic. Although users probably get the same impression, the very fact that system responded to their utterance with a joke apparently triggered them to evaluate it higher.

This leads us to another interesting point of view on the subject – does the knowledge that the partner (in user's evaluation) or one of the speakers (in third person evaluation) is a computer system influence the results? In case when evaluators know that, do they admire the system for trying to tell jokes, or rather dislike it for doing it less swiftly than humans? The results of user's evaluation (question F) showed, that "humanlikeness" of the system increased - however, this issue still requires more investigation.

In this study, in the third person evaluation experiment, the participants were not told of the origin of the dialogues. Therefore, there is a need of conducting a similar experiment, in which the participants would know that the dialogues took place between humans and computer systems.

7 What Solutions Do We Propose?

The overall message coming from this experiment is that research on humor processing is a worthwhile enterprise and is definitely worth continuing. However, of course, many improvements are still to be made, in talking system itself as well as in pun generator. Also, new evaluation methods will have to be considered, as assessing the "quality" of humor is difficult even in case of humans. Some ideas on these subjects are discussed below.

7.1 Solution to the Timing Problem

One of important problems in humor processing is the timing of jokes. Obviously, it is not always appropriate to tell a pun – for example, when the user is angry, a badly

timed joke may only boost his anger. One solution to the timing problem would be an implementation of an emotive analysis system, that would detect the user's emotional states and decide, whether it is proper to tell a pun. At this point of research, we are planning to use Ptaszynski's MLAsk Emotive Analysis System [13] and combine it with PUNDA's algorithm.

7.2 Evaluation Methods

Evaluation of joking systems is a serious problem in the field of humor processing. Even if we construct a perfect set of most subject-relevant questions, after all, it is humor that we assess – and this, by definition, can not be completely objective. However, some methods may prove useful as far as joking conversational systems are concerned. We are considering conducting of humor-oriented Turing Test [14], in which evaluators will have no knowledge of who their conversation partner is, and afterwards will have to guess, if it was a human or a machine. The experiment would include a comparison between humorous and non-humorous systems. This method, albeit not perfect, will presumably give credible results for the humanlikeness category of system's evaluation.

8 Conclusion

In this paper, we proved that implementation of a joke generator into a non-task oriented conversational system enhances its performance. Humor-equipped system was evaluated higher than the non-humor one, by the users as well as third person evaluators. We discussed the results, pointed out some problems and proposed their possible solutions. However, what is more important, the overall results of these experiments show, that combining the field of humor processing with research project on freely talking systems is a step in the right direction and under no circumstances should not be abandoned in the future.

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References

1. Morkes, J., Kernal, H.K., Nass, C.: Effects of humor in task-oriented human-computer interaction and computer-mediated communication: A direct test of srct theory. Hum-Comput. Interact. 14(4), 395–435 (1999)
2. Reeves, B., Nass, C.: The media equation: How people treat computers, television, and new media like real people and places. Cambridge Univ. Press, Cambridge (1996)
3. Binsted, K.: Using humour to make natural language interfaces more friendly. In: Proceedings of the AI, ALife and Entertainment Workshop, Intern. Joint Conf. on Artificial Intelligence (1995)
4. Binsted, K.: Machine humour: An implemented model of puns. Univ. of Edinburgh (1996)

5. Loehr, D.: An integration of a pun generator with a natural language robot. In: Hulstijn, J., Nijholt, A. (eds.) Proc. Intern. Workshop on Computational Humor, pp. 161–172. University of Twente, Netherlands (1996)
6. Dybala, P., Ptaszynski, M., Rzepka, R., Araki, K.: Extracting Dajare Candidates from the Web - Japanese Puns Generating System as a Part of Humor Processing Research. In: Proceedings of LIBM 2008, pp. 46–51 (2008)
7. Higuchi, S., Rzepka, R., Araki, K.: A casual Conversation System Using Modality and Word Associations Retrieved from the Web. In: Proceedings of EMNLP, Waikiki, Hawaii (to appear, 2008)
8. Goo search engine, `http://www.goo.ne.jp/`
9. Kudo, T.: MeCab: Yet another part-of-speech and morphological analyzer (2001), `http://mecab.sourceforge.ne`
10. MeCab-skkserv Kanji-Kana converter, `http://chasen.org/~taku/software/mecab-skkserv/`
11. Yoshihira, K., Takeda, T., Sekine, S.: KWIC system for Web Documents (in Japanese). In: Proceedings of the 10th Annual Meetings of the Japanese Association for NLP, pp. 137–139 (2004)
12. Sjöbergh, J., Araki, K.: Robots Make Things Funnier. In: Proceedings of LIBM 2008, pp. 46–51 (2008)
13. Ptaszynski, M., Dybala, P., Rzepka, R., Araki, K.: Effective Analysis of Emotiveness in Utterances Based on Features of Lexical and non-Lexical Layer of Speech. In: Proceedings of NLP Conference, Tokyo (2008)
14. Turing, A.: Computing Machinery and Intelligence. Mind 59(236), 433–460 (1950)